

DSA0204: COMPUTER VISION & OPENCV

Assignment: Image Formation, Sampling, Quantization and Levels of Computer Vision

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Question 1: Sampling and Image Quality (Blur Correction)

How sampling affects image quality

Sampling is the process of converting a continuous, spatially varying scene into a discrete grid of pixels. The camera's sensor array measures light intensity only at fixed spatial locations, so the sampling rate (equivalently, the pixel density or spatial resolution) determines how much of the scene's fine detail is actually captured.

When the sampling rate is too low relative to the spatial frequency content of the scene, two related problems occur:

- Loss of high-frequency detail: fine edges, textures and small objects fall between sample points and cannot be reconstructed, which is perceived as blur or softness in the image.
- Aliasing: according to the Nyquist criterion, the sampling rate must be at least twice the highest spatial frequency present in the scene. If this is violated, high-frequency patterns fold back into lower frequencies, producing false patterns (moire, jagged edges) rather than a faithful representation.
- Interpolation error: when an image is later resized, undersampled data forces the interpolation algorithm to guess missing information, compounding the blurred appearance.

In short, insufficient sampling reduces the effective spatial bandwidth of the captured image, so detail that exists in the real scene is simply never recorded, and what is recorded is corrupted by aliasing.

Corrective method

The appropriate correction depends on whether the blur originates before or after sampling:

- Increase the sampling rate: use a sensor with higher pixel density (more megapixels) or move closer to the subject, so that the pixel pitch is small enough to satisfy the Nyquist criterion for the finest detail of interest.
- Apply an anti-aliasing (low-pass) filter before sampling: an optical low-pass filter or a slight, controlled optical blur removes spatial frequencies above the Nyquist limit before they are sampled, trading a small amount of sharpness for the elimination of aliasing artifacts.

- Post-capture restoration: if the blur is due to a known point spread function (e.g., defocus or motion blur introduced during acquisition), deconvolution or Wiener filtering can partially restore lost detail.
- Super-resolution reconstruction: combine multiple slightly shifted low-resolution frames of the same scene, or use a trained super-resolution model, to synthesize a higher-resolution, sharper estimate of the original scene.

For a genuinely undersampled single capture, increasing sensor resolution or applying super-resolution is the most effective corrective approach, since no post-processing can recreate information that was never sampled in the first place.

Question 2: Image Sensing and Acquisition Under Low Light

The final digital image is the end product of a chain: illumination -> scene reflectance -> optics -> sensor -> analog-to-digital conversion. Under low-light conditions, each stage of this chain is stressed, and the weaknesses compound in the final image.

Stage-by-stage influence

- Illumination and reflectance: with little incident light, the amount of light reflected from scene surfaces toward the camera is small, so the signal reaching the sensor is inherently weak.
- Optical acquisition: the lens aperture may need to widen (lower f-number) to admit more light, which reduces depth of field, and exposure time is often increased, which risks motion blur for moving subjects or hand-shake.
- Sensor response: photosites convert incoming photons to electrons. In low light, few photons strike each photosite, so the signal is small relative to the sensor's intrinsic noise sources: photon shot noise (statistical fluctuation in photon arrival), dark current noise (thermally generated electrons), and read noise from the readout electronics.
- Gain/ISO amplification: to compensate for a weak signal, the sensor's analog gain (ISO) is increased. This amplifies the wanted signal, but it amplifies the noise floor proportionally, so the visible graininess in low-light photos increases.
- Quantization: the amplified analog signal is digitized into a fixed number of intensity levels. Because the useful signal range is compressed near the dark end, fewer quantization levels are actually used to represent the scene, increasing banding and reducing tonal gradation.

Net effect on the final digital image

The final image typically shows reduced signal-to-noise ratio, visible grain/speckle, compressed dynamic range, possible color shifts (as different colour channels are amplified unevenly), and sometimes motion blur from the longer exposure. Understanding this chain is what allows targeted correction: better optics and larger photosites improve the physical signal, while software techniques (multi-frame averaging, denoising, and adaptive gain) address the noise introduced at the sensing and amplification stages.

Question 3: Aliasing Artifacts: Cause and Correction

Cause of aliasing

Aliasing is fundamentally a sampling problem, not a quantization problem. It occurs when the spatial sampling rate of the imaging system is lower than twice the highest spatial frequency present in the scene, violating the Nyquist-Shannon sampling theorem. High-frequency detail (fine repetitive patterns such as fabric weaves, window blinds, or brick patterns) is then indistinguishable from a lower frequency, and the reconstructed image displays spurious patterns, such as moire fringes, jagged (staircase) edges, or false colour bands, that were never present in the original scene.

Quantization, by contrast, concerns how many discrete intensity levels are used to represent brightness at each already-sampled point. Coarse quantization produces a different artifact known as false contouring or banding, where smooth intensity gradients appear as visible steps. It is important to distinguish the two: aliasing is a spatial-frequency (sampling) artifact, while contouring is an amplitude-resolution (quantization) artifact.

Corrective approach

- Anti-aliasing (low-pass) filtering before sampling: attenuate spatial frequencies above the Nyquist limit optically or electronically prior to sampling, so no frequency content exists that could fold back and alias.
- Increase the sampling rate: use a higher-resolution sensor so the Nyquist frequency rises above the highest frequency content in the scene.
- Supersampling/oversampling: capture or render at a higher resolution than needed and then downsample using a proper low-pass filter, which is standard practice in both digital cameras and computer graphics.
- Software mitigation: apply a mild Gaussian smoothing before any resizing operation, and use high-quality resampling kernels (bicubic/Lanczos) instead of nearest-neighbour when resizing images.

Correct quantization (adequate bit depth, e.g., 8 bits or higher per channel) should also be ensured, since although it does not cause aliasing, insufficient bit depth combined with aliasing artifacts makes the overall degradation more visually severe.

Question 4: Pixel Resolution vs Intensity Resolution in Medical Imaging

Digital images are characterised along two independent axes: pixel resolution (spatial resolution), the number of sample points used to represent the scene, and intensity resolution (radiometric/bit-depth resolution), the number of distinct grey or colour levels available at each sample point.

Aspect	Pixel (Spatial) Resolution	Intensity (Amplitude) Resolution
Definition	Number of pixels (rows x columns) sampling the scene	Number of discrete intensity levels per pixel (bit depth)
Determines	Fineness of spatial detail, ability to resolve small structures	Ability to distinguish subtle differences in brightness/tissue density
Too low causes	Blocky images, loss of fine anatomical boundaries, missed small lesions	Banding/false contouring, loss of subtle contrast between similar tissues
Typical medical value	512 x 512 or higher for CT/MRI slices	12-16 bits per pixel (4096-65536 levels) for CT/MRI/DICOM
Cost of increasing	Larger file size, longer acquisition/reconstruction time	Larger file size, more sensitive to sensor/electronic noise

Practical implication in medical imaging

In modalities such as CT, MRI and digital X-ray, both resolutions matter for different clinical purposes. High pixel resolution is essential to delineate the precise shape and boundary of small anatomical structures, for example a small tumour margin or a hairline fracture, since a structure smaller than a few pixels cannot be reliably identified. High intensity resolution is essential because many diagnostically relevant differences (for example between healthy and diseased soft tissue) correspond to very small differences in absorption or signal intensity that would be indistinguishable if compressed into too few grey levels. This is why medical images are commonly stored at 12 to 16 bits per pixel even though they are typically displayed to the human eye, which only distinguishes around 6-7 bits of grey levels at once, because the extra bit depth allows radiologists to apply windowing (contrast stretching) to different intensity ranges without introducing banding.

Question 5: Classification of Computer Vision Level for Real-Time Surveillance

Levels of Computer Vision

Level	Typical Tasks	Role in Surveillance
Low-level	Image acquisition, noise reduction, contrast enhancement, edge detection	Continuous frame capture, denoising, illumination correction from the live video feed
Mid-level	Segmentation, feature extraction, object detection and tracking	Detecting and tracking people/vehicles frame to frame across the video stream
High-level	Scene understanding, event/behaviour recognition, decision making	Recognising an intrusion, loitering, or abandoned object and triggering an alert

Classification and justification

A real-time surveillance system spans all three levels in a continuous pipeline, but its defining, distinguishing capability sits at the mid-to-high level of Computer Vision. This is because the value of a surveillance system is not merely capturing clean frames (low-level), but its ability to detect, track and classify moving objects in real time (mid-level) and then interpret those detections as meaningful events, such as unauthorised entry or abnormal behaviour, and act on them (high-level).

The real-time constraint reinforces this classification: mid-level tasks such as object detection and tracking must run at video frame rate (commonly 15-30 fps), which drives the choice of lightweight detection/tracking algorithms, while high-level decision logic must operate with low latency to be operationally useful (for example, to trigger an alarm before an intruder leaves the frame). Low-level processing is necessary but is treated as a fast, largely automatic pre-processing stage rather than the differentiating capability of the system.

Question 6: Image Formation Models in Autonomous Driving

An image formation model mathematically describes how a 3D scene is transformed into a 2D image, typically combining a geometric model, a radiometric model, and a sensor model.

- Geometric model: describes the perspective projection of 3D world points onto the 2D image plane through the camera's intrinsic parameters (focal length, principal point, lens distortion) and extrinsic parameters (camera position and orientation).
- Radiometric model: describes how scene illumination and surface reflectance properties (BRDF) combine to determine the brightness recorded at each image point.
- Sensor/noise model: describes how the sensor converts incident light into a digital signal, including quantisation and noise characteristics.

How this improves image quality and perception in autonomous driving

- Lens distortion correction: knowing the precise geometric model (from camera calibration) allows radial and tangential lens distortion to be removed, so that straight lines in the world appear straight in the image, which is critical for accurate lane and object localisation.
- Depth and distance estimation: the geometric model underlies stereo vision and structure-from-motion, allowing the vehicle to compute the distance to obstacles, pedestrians and other vehicles from 2D image correspondences.
- Robustness to illumination: the radiometric model helps separate the effects of changing illumination (sunlight, shadows, headlights, tunnels) from genuine object appearance, improving detection consistency across lighting conditions.
- Noise-aware processing: an accurate sensor noise model allows perception algorithms to apply appropriately tuned filtering, so that real obstacles are not confused with sensor noise, and genuine faint signals are not filtered away.
- Sensor fusion and calibration: since autonomous vehicles combine cameras with LiDAR and radar, precise geometric image formation models are essential to align (calibrate) these different modalities into a single consistent 3D understanding of the environment.

Overall, the image formation model provides the mathematical foundation needed to correct, calibrate, and interpret raw camera data so that downstream perception modules (lane detection, object detection, depth estimation) receive geometrically accurate and radiometrically consistent images, directly improving the safety and reliability of the driving system.

Question 7: Role of Quantization in Grayscale Image Digitization

Quantization is the process that assigns each sampled point of a continuous-tone grayscale image to one of a finite set of discrete intensity (grey) levels, so that the image can be represented and stored digitally with a finite number of bits per pixel.

Role in digitization

- Converts a continuous amplitude range (in principle infinite possible brightness values) into a fixed number of representable levels, commonly $L = 2^k$ levels for a k-bit image (for example, 256 levels for an 8-bit image).
- Enables digital storage, transmission and processing, since computers can only represent numbers with finite precision.
- Works together with spatial sampling: sampling fixes where measurements are taken, while quantization fixes what values those measurements are allowed to take.

Impact on image detail

- Coarse quantization (few bits, e.g., 1-4 bits): produces visible false contouring/banding, where smooth intensity gradients (like a sky or a shadow gradient) are rendered as abrupt, visible steps rather than smooth transitions, and subtle low-contrast detail is lost entirely.
- Fine quantization (e.g., 8 bits or more): allows smooth tonal gradation and preserves subtle brightness differences, at the cost of increased storage per pixel.
- Trade-off: increasing the number of quantization levels improves perceived image quality and detail but increases file size and computational cost; below roughly 6 bits (64 levels), most images begin to show noticeable banding to the human eye, which is why 8-bit (256-level) grayscale is the common practical standard.

In short, quantization directly controls the tonal fidelity of the digitized image: too few levels sacrifice fine intensity detail for compactness, while sufficient levels preserve the perceptual smoothness of the original continuous-tone scene.

Question 8: Noise in Image Acquisition and Its Minimization

How the acquisition process contributes to noise

- Photon shot noise: light itself arrives at the sensor as discrete photons following Poisson statistics, so even a perfect sensor exhibits statistical fluctuation in photon count, which is more significant relative to signal strength at low light levels.
- Dark current noise: thermally generated electrons accumulate in the sensor even without incoming light, and this thermal noise increases with sensor temperature and exposure time.
- Read noise: the electronics that convert and amplify the sensor's charge into a digital voltage introduce noise during readout, which is roughly constant regardless of signal level.
- Amplifier (gain/ISO) noise: increasing the analog gain to compensate for low signal amplifies the noise sources above along with the desired signal.
- Quantization noise: the analog-to-digital conversion itself introduces a small rounding error at each sample, which becomes more significant when the meaningful signal only occupies a small part of the available intensity range.

How noise can be minimized

- Improve the physical acquisition conditions: increase scene illumination, use a larger sensor or larger photosites (which collect more photons per pixel), or use a longer exposure where motion allows, all of which raise the signal-to-noise ratio at the source.
- Sensor cooling: reducing sensor temperature (used in scientific and astronomical imaging) reduces dark current noise.
- Multi-frame averaging: capturing several frames of a static or aligned scene and averaging them reduces random (shot and read) noise, since random noise partially cancels while the true signal reinforces.
- Software denoising: apply spatial filters such as Gaussian, median, bilateral or non-local means filtering, or learned denoising models, to suppress noise while attempting to preserve edges and detail.
- Careful gain management: avoid unnecessarily high ISO/gain settings, and use noise-aware image signal processing pipelines that apply stronger denoising specifically in low-signal regions.

Effective noise reduction generally combines improving the physical acquisition (more light, better sensor, appropriate exposure) with post-acquisition software denoising, since no single stage alone eliminates all noise sources described above.

Question 9: Levels of Computer Vision in Facial Recognition

Facial recognition is a good illustration of how the classical three-tier hierarchy of Computer Vision, low-level, mid-level and high-level, work together as a pipeline from raw image acquisition to a final identity decision.

Level	Facial Recognition Stage	Description
Low-level	Image acquisition and pre-processing	Camera captures the face image; pre-processing performs illumination correction, contrast normalisation, denoising and geometric alignment/cropping of the face region.
Mid-level	Feature extraction and representation	Detection of the face region, localisation of facial landmarks (eyes, nose, mouth), and extraction of discriminative features (e.g., via a convolutional embedding network) that represent the face compactly.
High-level	Recognition and decision making	Comparison of the extracted feature embedding against a database of known identities (matching/classification), applying a decision threshold, and outputting an identity or an accept/reject/liveness decision.

Why all three levels are necessary

Each level depends on the quality of the one before it: poor low-level pre-processing (for example, harsh shadows or noise left uncorrected) degrades the accuracy of mid-level landmark detection and feature extraction, which in turn reduces the reliability of the high-level identity decision. Conversely, sophisticated high-level matching algorithms cannot compensate for a poorly acquired or poorly aligned input face image. This is why practical facial recognition systems are engineered as an end-to-end pipeline spanning all three levels, from raw pixel acquisition through geometric and photometric normalisation, to feature-level representation, and finally to a semantic, high-level identity decision.

Question 10: Effect of Low Resolution on Object Visibility

Role of sampling

Pixel resolution determines how densely the continuous scene is sampled onto the sensor grid. When resolution is low, each pixel integrates light from a comparatively large area of the scene, so an object that would otherwise span many pixels at higher resolution may span only a handful of pixels, or even less than one pixel, at low resolution. Fine structural detail (edges, corners, texture) that a detection algorithm relies on to distinguish an object from its background is therefore averaged away or lost entirely, exactly as described by the sampling and Nyquist arguments in Questions 1 and 3.

Role of image formation

The image formation process projects the 3D scene through the lens onto a finite sensor array; a smaller number of sensor elements (lower resolution) means the geometric projection is coarsely discretised, so the apparent size, shape and boundary of small or distant objects become poorly defined or ambiguous. Combined with the point spread function of the optical system, low-resolution acquisition effectively low-pass filters the scene, blurring the sharp intensity transitions (edges) that most object-detection features (gradients, contours, keypoints) depend on.

Combined effect on detection

- Small or distant objects may be represented by too few pixels to preserve a distinctive shape, so their silhouette becomes indistinguishable from background clutter.
- Edge and gradient-based features used by classical and deep-learning detectors become weak or noisy, lowering detection confidence and increasing false negatives/positives.
- Aliasing introduced by undersampling (Question 3) can further distort the object's apparent shape, misleading shape-based recognition.

Corrective directions

- Use a higher-resolution sensor or move the camera/lens closer (larger focal length) so the object of interest occupies more pixels.
- Apply super-resolution reconstruction to enhance an already-captured low-resolution image before running detection.
- Ensure adequate anti-aliasing/optical quality during acquisition so that the limited pixels that are captured are not further corrupted by aliasing.

In summary, both the sampling density and the geometric/optical image formation process jointly determine whether an object retains enough distinguishable spatial structure to be reliably detected; when resolution is too low, objects are effectively undersampled below the level of detail required by detection algorithms, directly causing failed or unreliable detection.